

01_MASTER_PROJECT_BLUEPRINT.md

SIH26034 - Legal Metrology Automated Compliance System

The Master Single Source of Truth (SSOT) & Architectural Blueprint

1. Executive Summary & Project Identity

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PROJECT IDENTIFIER	: SIH26034
PROJECT TITLE	: Software System to check compliance of Packaged Commodities under
	Legal Metrology (Packaged Commodities) Rules, 2011 by scanning products, images and labels.
MINISTRY	: Ministry of Consumer Affairs, Food & Public Distribution
DEPARTMENT	: Department of Consumer Affairs (DoCA)
COMPETITION	: Smart India Hackathon 2026
ENGINEERING TEAM	: 6-Member Multidisciplinary Student Engineering Team
CURRENT STAGE	: FINAL ARCHITECTURE FREEZE COMPLETED -> READY FOR DEVELOPMENT
TARGET DEADLINE	: 13 September 2026 (6 Days Execution Window)
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This document serves as the **Master Single Source of Truth (SSOT)** for project SIH26034. It unifies all domain research, legal statutory standards, optical mathematics, machine learning pipelines, user experience designs, security architectures, and execution workstreams into an authoritative, binding engineering blueprint.

2. Complete Repository Blueprint & Documentation Map

The repository contains 18 authoritative engineering and governance specifications:

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FILE NAME	DESCRIPTION & PRIMARY SCOPE
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01_MASTER_PROJECT_BLUEPRINT.md	Master SSOT synthesizing all research, architecture, & plan.
02_FINAL_REQUIREMENTS_SPECIFICATION.md	Master Requirements Spec (FR-01 to FR-20, NFR-01 to NFR-10).
03_FINAL_ARCHITECTURE.md	Modular monolith architecture & 12-stage pipeline data flow.
04_FINAL_MVP_SCOPE.md	MoSCoW prioritization, P0 freeze, & explicit cut list.
05_TECHNOLOGY_DECISION_RECORD.md	Formal ADRs (ADR-01 to ADR-12) covering stack choices.
06_DATA_AND_MODEL_STRATEGY.md	DS-SYNTH-001, DS-PILOT-050, ONNX INT8, & model licensing.
07_API_AND_INTERFACE_CONTRACTS.md	RESTful OpenAPI 3.1 endpoints, Pydantic v2 pipeline DTOs.
08_DATABASE_SPECIFICATION.md	Relational DDL, SQLite/PostgreSQL schemas, & index strategy.
09_UI_UX_BLUEPRINT.md	Design tokens, mobile HUD, adjudication canvas, & notices.
10_SECURITY_AND_AUDIT_SPECIFICATION.md	RBAC matrix, Section 63 BSA 2023, & Merkle audit ledger.
11_TESTING_AND_VALIDATION_PLAN.md	4-tier test pyramid, 20 formal test suites, & acceptance QA.
12_DEMO_PLAN.md	3-tier presentation fallback architecture & pitch script.
13_SIX_MEMBER_EXECUTION_PLAN.md	Granular workstreams, APIs, & day-by-day tasks for 6 devs.
14_GITHUB_WORKFLOW.md	Trunk-based branching, PR policy, CI checks, & DoD.
15_RISK_AND_CONTINGENCY_REGISTER.md	10 red-team failure modes, tripwires, & mitigation plans.
16_DECISION_LOG.md	Architectural Decision Log (ADL-01 to ADL-12) & rationales.
17_OPEN_QUESTIONS.md	Strictly bounded open questions with frozen working defaults.
CLAIMS_WE_MUST_NOT_MAKE.md	Blacklist of unsupported, legally hazardous claims.
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3. Master Synthesis of All Three Research Phases

3.1 Phase 1: Legal & Domain Metrology Foundation

- **Statutory Authority:** Enforcement is rooted in the *Legal Metrology Act, 2009* and the *Legal Metrology (Packaged Commodities) Rules, 2011*.
- **Critical Gazette Corrections:**
 - *Table-I Font Height:* G.S.R. 629(E) dated 23.06.2017 establishes that for packaging areas $> 2500 \text{ cm}^2$, the minimum numeral height is **6.0 mm** (correcting historical typos citing 8.0 mm).
 - *Unit Sale Price (USP):* G.S.R. 779(E) dated 02.11.2021 mandates explicit USP declarations (per gram / per milliliter for $< 1\text{kg}/1\text{L}$, and per kg / per L for $> 1\text{kg}/1\text{L}$) rounded to 2 decimal places.
 - *E-Commerce Mandates:* Rule 6(10) mandates digital declaration of all Rule 6 attributes on e-commerce platforms. G.S.R. 128(E) dated 13.02.2026 inserts Rule 6(10A) mandating structured, searchable Country of Origin filters effective 1 July 2026.
 - *Electronic Evidence Repeal & Replacement:* The Indian Evidence Act, 1872 (Section 65B) was repealed on 1 July 2024. All digital evidence admissibility is governed exclusively by **Section 63 of the Bharatiya Sakshya Adhiniyam, 2023 (BSA 2023)**.

3.2 Phase 2: Computer Vision, Optical Mathematics & Model Strategy

- **Physical Scale Solvability:** Uncalibrated monocular depth estimation suffers from scale ambiguity and cannot measure physical millimeters. The system enforces **Planar Homography with OpenCV ArUco 4x4 fiducial markers** (with fallback to ₹5 standard coins), achieving $\leq 0.30 \text{ mm}$ error margin.
- **Licensing Safety:** Ultralytics YOLOv8 and YOLOv11 are strictly excluded due to GNU AGPL-3.0 viral copyleft terms. The system is built 100% on permissive **Apache-2.0 / MIT** models:
 - Text Detection: DBNet++ (Real-time polygon detector)
 - Text Recognition: PaddleOCR PP-OCRv4
 - Panel Segmentation: RT-DETR / OpenCV classical contours
- **Edge Inference:** ONNX Runtime INT8 quantization enables sub-second ($< 1200 \text{ ms}$) execution on standard 8-core CPUs without GPU hardware.

3.3 Phase 3: System Engineering, Human Adjudication & Evidence Architecture

- **Advisory Diagnostic AI:** Under Indian constitutional jurisprudence, AI cannot act as an autonomous magistrate or automatically penalize citizens. The system serves as an **Augmented Diagnostic Assistant**; all adverse findings pass through a mandatory **Human-in-the-Loop (HITL) Inspector Adjudication Canvas**.
- **Cryptographic Chain of Custody:** Images are hashed with SHA-256 upon ingestion and recorded in an append-only Merkle ledger. All statutory notices under Section 36(1) are exported as archival PDF/A documents accompanied by Section 63 BSA 2023 Electronic Evidence Certificates.

4. Frozen Core Facts, Verified Boundaries & Technical Constants

+-----+ -----+ PARAMETER / CONSTANT FROZEN VALUE SOURCE / AUTHORITY +-----+ -----+		
Maximum Acceptable Inference Latency (8-core CPU)	<= 1200 ms	System NFR-01
Minimum Image Resolution Quality Standard	1920 x 1080 (1080p)	Optical
Minimum Laplacian Blur Variance Specification	150.0	Quality Gate
Maximum Permissible Specular Glare Specification	3.0% of label area	Quality Gate
Maximum Permissible Camera Tilt Warp Bound	15.0 degrees	Homography
Primary Fiducial Standard 06	ArUco 4x4_50 (50.0 mm)	ADL-05 / ADR-
Secondary Fiducial Standard 06	Indian ₹5 Coin (23.0 mm)	ADL-05 / ADR-
Font Measurement Tolerance Benchmark Ground Truth	+/- 0.30 mm MAE	Caliper
Table-I Area > 2500 cm² Font Bound 629(E) 23.06.2017	6.0 mm (NOT 8.0 mm)	Gazette G.S.R.
Electronic Evidence Statute Sakshya Adhiniyam	Section 63 BSA 2023	Bharatiya
E-Commerce Origin Filter Mandate 128(E) 13.02.2026	Rule 6(10A)	Gazette G.S.R.
Legal Notice Statutory Format LM Act 2009	Form-1/ Form-2	Section 36(1)
Primary Database Engine (Local) 04	SQLite 3.45+ (SQLCipher)	ADL-04 / ADR-

Primary Database Engine (Cloud)	PostgreSQL 16+	ADL-04 / ADR-04
Primary Backend Framework	FastAPI 0.110+ (Py 3.11+)	ADR-01
Primary Frontend Framework	React 18+ (Vite + Tailwind)	ADR-02
Open-Source License Ceiling	Apache-2.0 / MIT / BSD	ADL-03 / ADR-03
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5. Day-by-Day Master Execution Schedule (07 - 13 September 2026)

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TIMELINE OVERVIEW:
- Day 1 (07 Sep): Project Setup, Repository Skeleton, DB Schemas, DTO Contracts.
- Day 2 (08 Sep): Synthetic Data Generation (DS-SYNTH-001) & Optical Pre-
Processing.
- Day 3 (09 Sep): OCR Integration (DBNet++ / PP-OCrv4) & Connected Component Font
Engine.
- Day 4 (10 Sep): Deterministic Rule Engine, HITL Adjudication UI & P0 Integration
Freeze.
- Day 5 (11 Sep): Section 63 BSA Notice PDF Generator, E-Commerce Auditor & P1
Polish.
- Day 6 (12 Sep): End-to-End Stress Testing, Caliper Benchmarking & Demo
Rehearsals.
- Day 7 (13 Sep): FINAL SUBMISSION & LIVE SMART INDIA HACKATHON EVALUATION.
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Detailed Daily Milestones

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DAY & DATE	WORKSTREAM MILESTONES & DELIVERABLES
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Day 1 (07 Sep)	M1: Setup GitHub repo, trunk rules, pre-commit hooks, CI.
(Foundations)	M2: Implement DB models (SQLite/Postgres), Alembic migrations.
	M3: Define Pydantic v2 DTOs, FastAPI skeleton routes.
	M4: Initialize React 18 + Vite + Tailwind scaffolding.

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| M5: Build synthetic label script & procure 50 FMCG pilot
packages. |
| M6: Setup automated test harness (pytest, Vitest, fixtures).
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| Day 2 (08 Sep) | M1: Build OpenCV ArUco detector & homography warp pipeline.
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| (Pre-Processing) | M2: Implement Laplacian blur & specular glare quality gate.
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| M3: Implement JWT authentication, RBAC middleware, Merkle
ledger. |
| M4: Build mobile camera HUD component with ArUco alignment box.
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| M5: Generate 2,000 synthetic labels with ground truth
annotations. |
| M6: Author unit tests for quality gate and fiducial
calibration. |
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| Day 3 (09 Sep) | M1: Integrate DBNet++ ONNX INT8 text detection pipeline.
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| (Vision & Font) | M2: Integrate PaddleOCR PP-OCRv4 text recognition pipeline.
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| M3: Build physical font x-height connected component
measurement. |
| M4: Implement interactive Adjudication Canvas (Pan/Zoom/Boxes).
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| M5: Benchmark OCR on DS-PILOT-050; tune character confidence.
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| M6: Author integration tests for OCR and font measurement.
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| Day 4 (10 Sep) | M1: Build deterministic Legal Rule Engine (Rule 6, Table-I,
USP). |
| (P0 FREEZE) | M2: Connect FastAPI pipeline end-to-end (Stage 1 to Stage 10).
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| M3: Wire frontend Canvas to live backend inspection API.
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| M4: Build Inspector override & justification remarks modal.
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| M5: Execute caliper benchmark: verify font MAE <= 0.30 mm.
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| M6: P0 FEATURE COMPLETE & CODE FREEZE. Execute full regression.
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| Day 5 (11 Sep) | M1: Implement ReportLab Form-1/2 Legal Notice PDF/A generator.
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| (Notices & ECom) | M2: Implement Section 63 BSA 2023 Digital Certificate
generator. |
| M3: Build E-Commerce single listing URL / DOM / gallery
auditor. |
| M4: Build Notice preview & download modal in frontend.
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| M5: Build Executive Dashboard KPI widgets & batch CSV queue
(P1). |
| M6: End-to-end integration test of notice generation and BSA

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certs. |
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| Day 6 (12 Sep) | ALL MEMBERS:
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| (Stress & Demo) | 1. Execute full 20-suite test pyramid across all 50 pilot
packages. |
|                | 2. Record 1080p Virtual Video Stream backup (Demo Tier 2).
|
|                | 3. Generate static evidence pack (Demo Tier 3).
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|                | 4. Conduct 5 dry-run presentations adhering to 3-minute pitch
script|
|                | 5. Verify zero lingering lints, warnings, or debug print
statements.|
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| Day 7 (13 Sep) | SUBMISSION & LIVE EVALUATION GATE:
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| (Evaluation)   | Deploy to demo laptops, verify offline mode, win Hackathon.
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6. Final Development Readiness Gate & Pre-Flight Checklist

Before launching implementation, the engineering team must satisfy the following **Pre-Flight Gates**:

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[x] 1. Legal Alignment: Table-I font boundary confirmed at 6.0 mm for >2500 cm²
(G.S.R. 629(E)).
[x] 2. Evidentiary Law: Section 63 BSA 2023 replaces repealed Section 65B Indian
Evidence Act.
[x] 3. E-Commerce Scope: Mass scraping rejected; single URL / DOM / image
inspection frozen.
[x] 4. Licensing Safety: AGPL-3.0 banned; DBNet++, PP-OCrv4, RT-DETR selected under
Apache-2.0.
[x] 5. Hardware Decoupling: ONNX INT8 CPU inference eliminates GPU server
dependency.
[x] 6. Optical Grounding: ArUco / coin fiducial homography eliminates scale
ambiguity.
[x] 7. Constitutional Process: Advisory diagnostic assistant with mandatory HITL
adjudication.
[x] 8. Tamper-Evidence: SHA-256 chained Merkle audit ledger specified.
[x] 9. Execution Plan: 6-member workstreams with explicit API interfaces and tasks
assigned.
[x] 10. Contingency: 3-tier demo fallback and 10 red-team risk mitigations
documented.
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STATUS: 10 / 10 GATES PASSED -> ARCHITECTURE FROZEN -> PROCEED TO DEVELOPMENT.